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Design and Implementation of a Population Growth Simulator

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1. Project Background

1.1. Domain Analysis and Current Solutions

The project is situated within the field of biology, specifically focusing on population dynamics. Population growth is a fundamental concept in ecology and has significant implications for conservation, resource management, and understanding ecosystem health.
While legacy solutions for this problem exist, they often suffer from limitations such as technical debt and they're not designed to be easily extendable. These solutions may not be able to adapt to new requirements or incorporate additional features without significant rework, which can hinder their long-term viability and usefulness.

1.2. Justification for Greenfield Development

This project is being developed as a greenfield project, meaning that it is being built from scratch without any constraints imposed by existing code or systems. This approach allows for greater flexibility and creativity in the design and implementation of the solution.
By starting with a clean slate, the project can be tailored to meet specific requirements and can incorporate modern technologies and best practices without being hindered by legacy code. This can lead to a more efficient development process and a more robust and maintainable solution in the long run.

1.3. Proposed Solution

2. Project Scope

2.1. Project Statement

The project aims to develop a population growth simulator that models the dynamics of population growth under various conditions, including sources of stochasticity. The simulator will allow users to input parameters such as initial population size, growth rate, carrying capacity, and time steps to visualize how populations evolve over time. The goal is to create a tool that can be used for educational purposes, research, and to gain insights into population dynamics.

2.2. System Boundaries

The proposed system is a **standalone** application. It does not require synchronization or integration with external legacy databases or third-party APIs for its core operation. All data persistence and business logic are managed internally.

2.3. Out of Scope

The project will not include features such as:
- Real-time data synchronization with external sources.
- Integration with third-party APIs for data retrieval or analysis.
- Advanced machine learning algorithms for predictive modeling of population growth.
- User authentication and multi-user support.